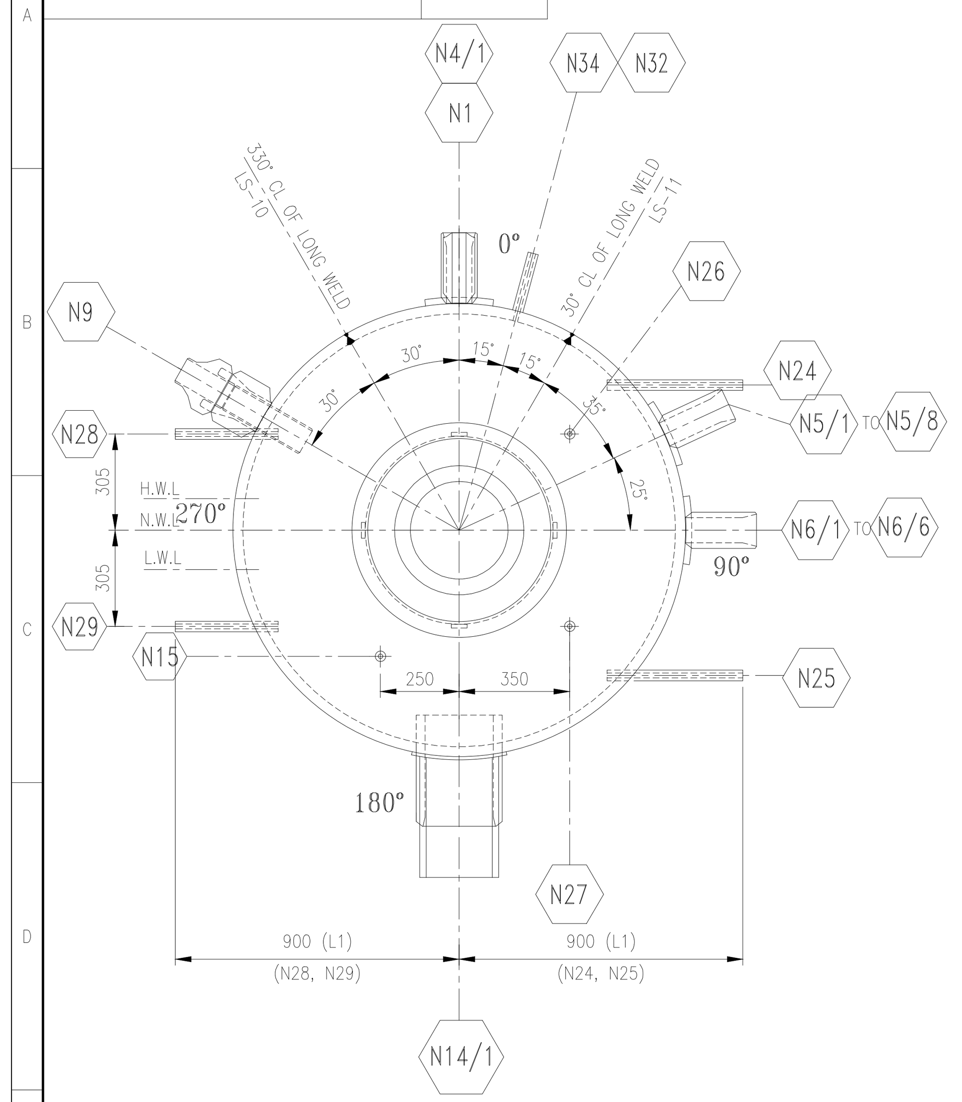
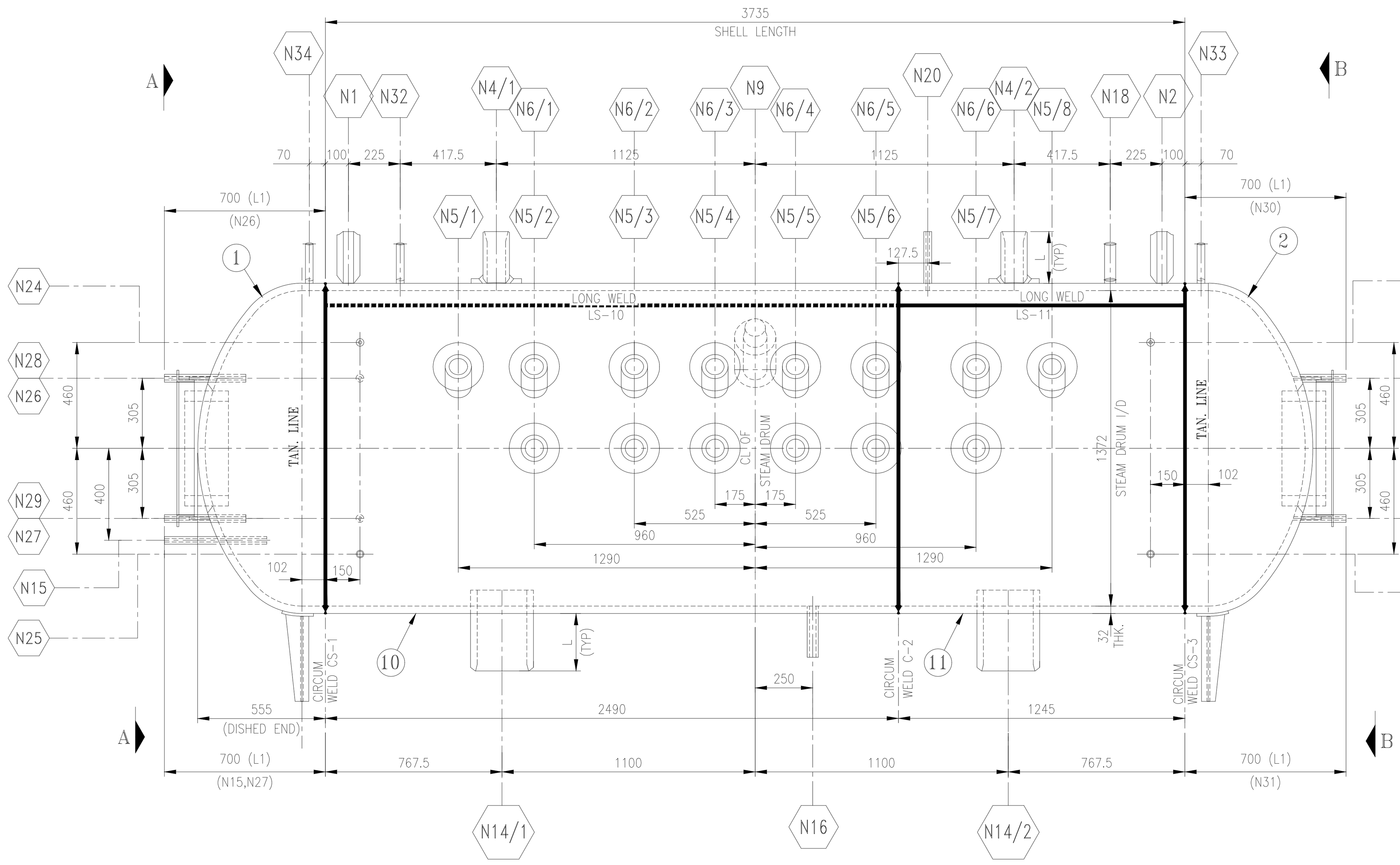
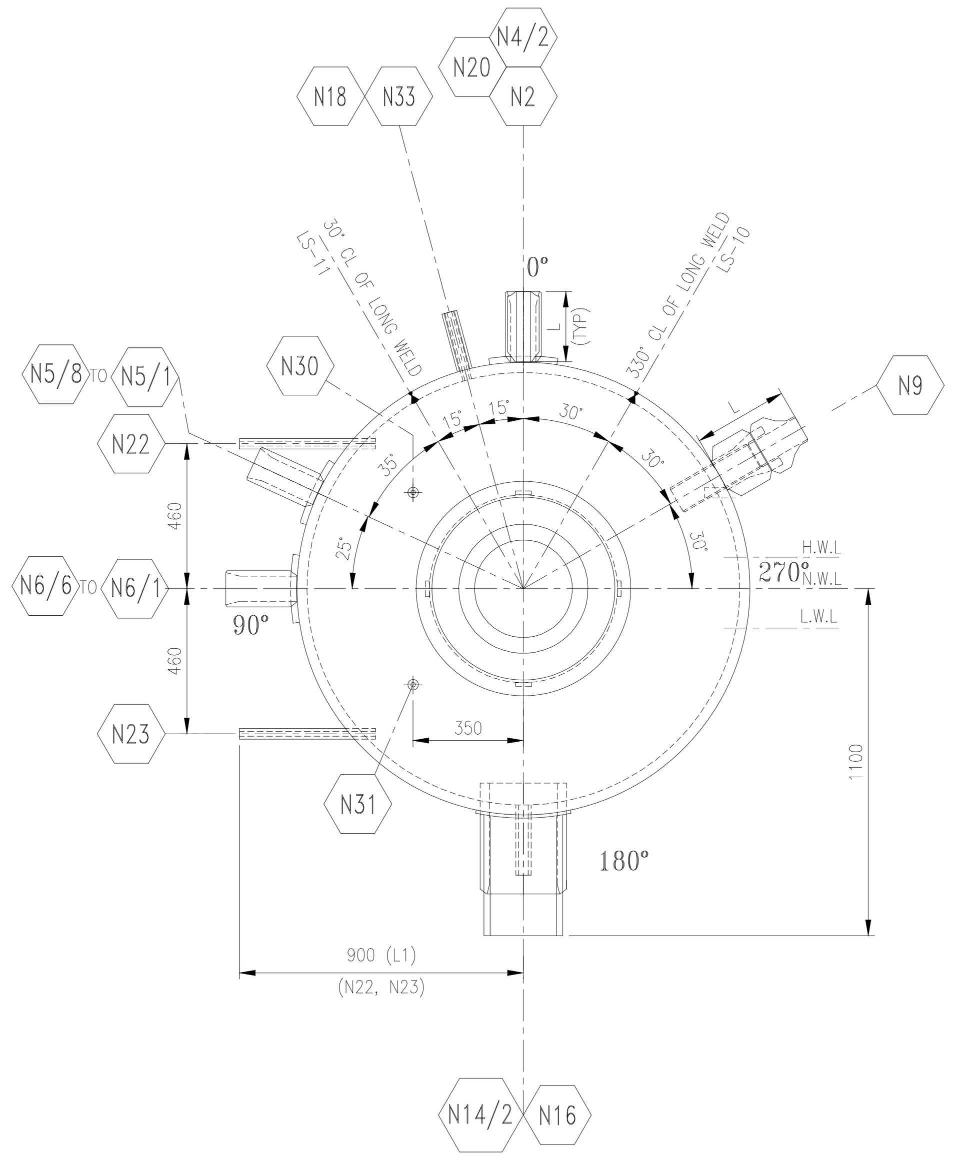
PROJECT ID	JB1160
DOCUMENT ID	JB1160-50010100-007-0001



VIEW - AA
(LHS OF BOILER)



ELEVATION (VIEWED FROM FRONT OF BOILER)
ARRANGEMENT OF STEAM DRUM



VIEW - BB
(RHS OF BOILER)

DESIGN DATA

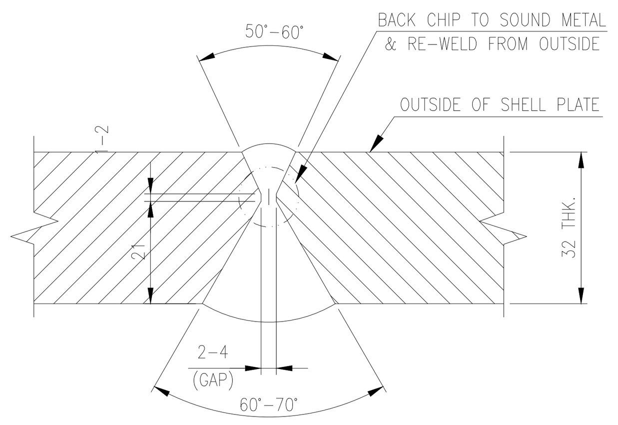
CODE OF CONSTRUCTION	IBR 1950 & ITS LATEST AMENDMENTS UP TO - SEPTEMBER 2019 (SEE NOTE: 02)
MAXIMUM WORKING PRESSURE	59 Kg/Sq. cm (g)
MAXIMUM WORKING TEMPERATURE	275 °C
WORKING METAL TEMPERATURE	275 °C
HYDRAULIC TEST PRESSURE	88.5 Kg/Sq. cm (g) (AT SHOP & AS PER IBR AT SITE)
RADIOGRAPHY OF LONG. & CIRCUM. SEAM	100%
STRESS RELIEVING	YES & REFER HEAT TREATMENT CYCLE TABLE
EVAPORATION OF BOILER (M.C.R.)	23000 Kg/Hr
TOTAL HEATING SURFACE AREA OF BOILER (EXCLUDING FORCED FLOW ELEMENT & SUPER HEATER)	940 Sq. m
HEATING SURFACE AREA OF FORCED FLOW SECTION (NON STEAMING)	575 Sq. m
STEAM PRESSURE AT SUPER HEATER OUTLET	45.0 Kg/Sq. cm (g)
SUPER HEATER OUTLET TEMPERATURE	400±5 °C
WEIGHT OF DRUMS (EMPTY)	7591.55 Kg. (APPROX.) (STEAM DRUM)
INSPECTION	CDB/LLOYD'S(SEE NOTE:3)

LIST OF REFERENCE DRAWINGS

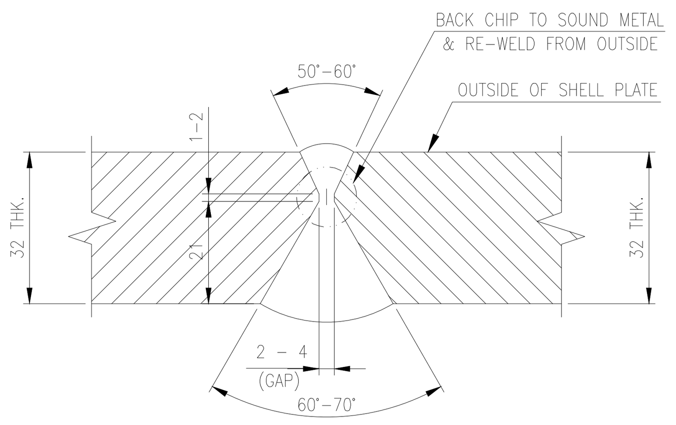
DRAWING DOCUMENT ID	DESCRIPTION
JB1160-50010100-006-0001	ARRANGEMENT & DETAIL OF CIRCULAR MANHOLE ON STEAM DRUM
JB1160-50010100-007-0002	DETAIL & MATERIAL LIST FOR STEAM DRUM
JB1160-50010100-009-0001	DETAIL OF DRUM NOZZLES

MATERIAL SPECIFICATIONS :

DESCRIPTION	SIZE	SPECIFICATION	MIN. TENSILITY (Kg/ Sq.mm)	MIN.CALC. THK. (mm)	REMARKS
SHELL BELTS	AS PER MATERIAL LIST	SA 516 Gr.70	49.45	30.36	FOR STEAM DRUM
DISHED ENDS	AS PER MATERIAL LIST	SA 516 Gr.70	49.45	24.71	FOR STEAM DRUM
LAGGING RING	AS PER MATERIAL LIST	IS 2062 Gr.A OR EQ.	-	-	FOR STEAM DRUM
CLEATS	AS PER MATERIAL LIST	IS 2062 Gr.A OR EQ.	-	-	FOR STEAM DRUM

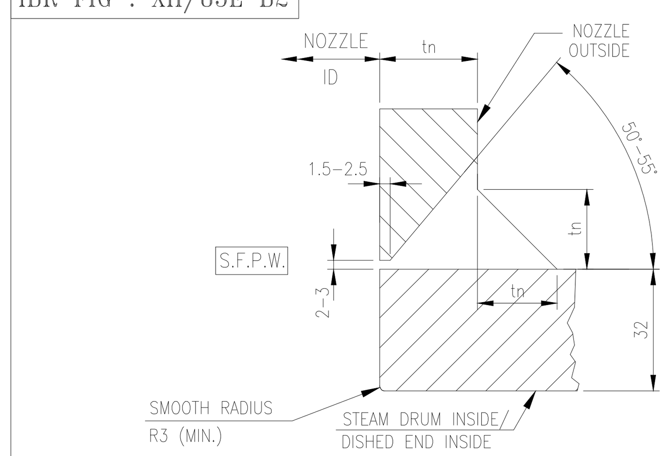


WELD PREPARATION OF LONG. SEAMS FOR STEAM DRUM
(SEAM MK. Nos. LS-10 & LS-11)



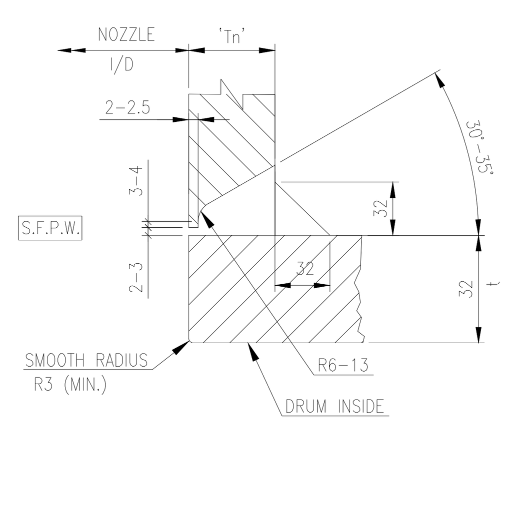
WELD PREPARATION OF CIRCUMFERENTIAL SEAMS FOR STEAM DRUM
(SEAM MK. Nos. CS-1, CS-2 & CS-3)

CHAPTER : XII
IBR FIG : XII/85E B2



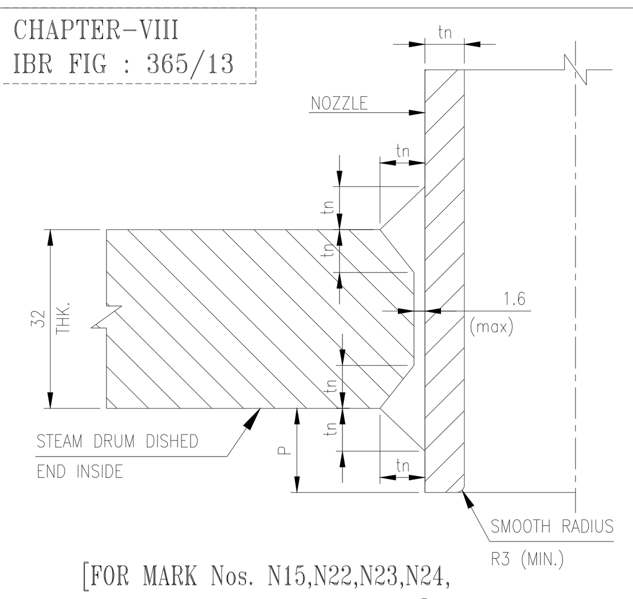
WELDING DETAIL OF NOZZLE TO DRUM
[FOR MARK Nos. N16,N18,N20,N32,N33,N34]

IBR CHAPTER - XII
IBR FIG : XII/85E J2



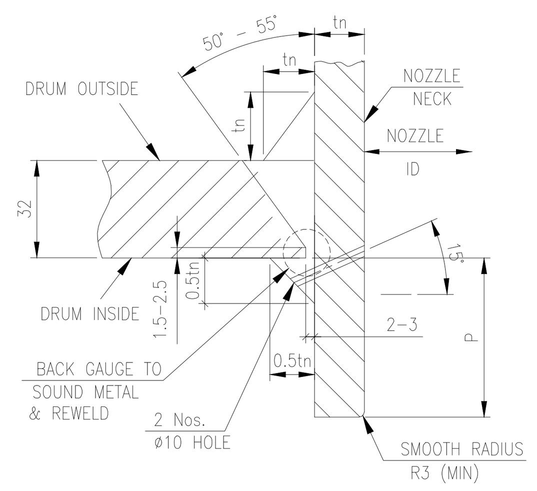
WELD DETAIL OF NOZZLE TO DRUM
MARK Nos. N9

CHAPTER-VIII
IBR FIG : 365/13



[FOR MARK Nos. N15,N22,N23,N24, N25,N26,N27,N28,N29,N30,N31]
WELDING DETAIL OF NOZZLE TO DRUM

CHAPTER - XII
IBR FIG : XII/85E B1



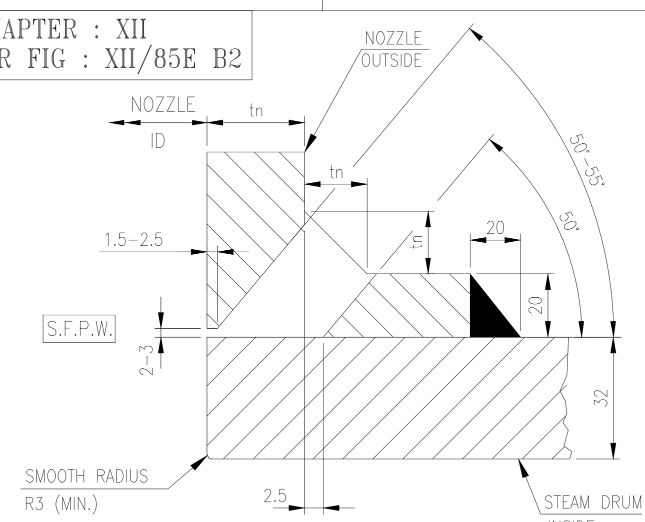
WELD DETAIL OF NOZZLE TO DRUM
MARK No. N14/1 & N14/2

NOTES :-

01. ALL DIMENSIONS ARE IN "mm" UNLESS OTHERWISE SPECIFIED.
02. THE ZERO DATE OF THIS CONTRACT IS 30.09.2019 . HENCE IBR AMENDMENTS UP TO SEPTEMBER 2019 HAVE BEEN TAKEN CARE OF WHILE DESIGNING THE EQUIPMENT.
03. INSPECTION BY DIRECTOR OF BOILERS, UTTAR PRADESH, BECAUSE MANUFACTURING OF THIS EQUIPMENT WILL BE DONE AT ATV PROJECT INDIA LIMITED, MATHURA, UTTAR PRADESH.
04. FOR GENERAL NOTES & MATERIAL SPECIFICATIONS REF. DOCUMENT ID : JB1160-50010100-007-0002.
05. MAXIMUM WORKING PRESSURE INDICATED IN DESIGN DATA ARE EXCLUSIVE OF STATIC HEAD.
06. THE SUPPORT ARRANGEMENT WILL NOT RESULT IN UNDUE STRESS BEING IMPOSED ON THE DRUM.
07. PWHT CYCLE :- STEAM DRUM
    - A) LOADING TEMP. (MAX.) = 300°C
    - B) RATE OF HEATING ABOVE 300°C (MAX.) = 100°C/Hr.
    - C) SOAKING TEMP. = 610°C ± 10°C
    - D) SOAKING TIME (MIN.) = 1 Hr 20 MINUTES.
    - E) RATE OF COOLING ABOVE 300°C (MAX.) = 100°C/Hr.
    - F) UNLOADING TEMP (MAX.) = 300°C
    - G) COOLING AFTER UNLOADING = COOLED IN STILL AIR

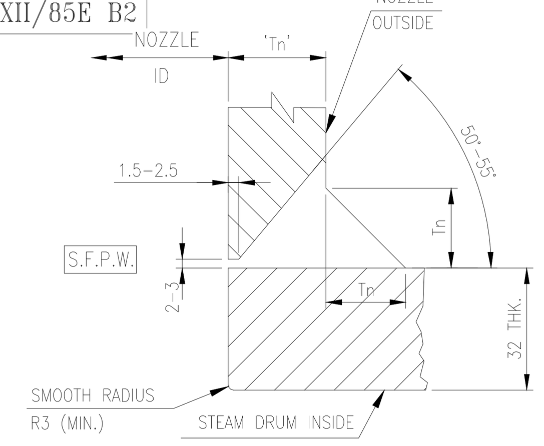
S.Nos.	Nozzle Nos.	Nozzle Descriptions	Location of Nozzle	No. of Nozzle	Nozzle Size (OD) (D) mm	Schedule of Nozzle	Nozzle Thickness (t) mm	Connecting Pipe Schedule/ Thickness (mm)	Projection from Drum OD (L) mm Drum Dished End Seam (L1) mm	Nozzle CL to Drum Vertical Axis (±X) mm	Nozzle CL to Drum Horizontal Axis (±Y) mm	Calculated Nozzle Length (R) mm	Finished Length of the Nozzle (M) mm	M/C Bore for end Preparation (E) mm	Weld Dim. (tn) mm	Nozzle Inside Projection(P) mm
1	N1	SAFETY VALVE NOZZLE-1	STEAM DRUM SHELL	1	100	-	28	-	225	-	-	255	225	44.45	28	0
2	N2	SAFETY VALVE NOZZLE-2	STEAM DRUM SHELL	1	100	-	28	-	225	-	-	255	225	44.45	28	0
3	N4/1 & N4/2	SATURATED STEAM STUB	STEAM DRUM SHELL	2	114.3	SCH.XXS	17.12	SCH.40	225	-	-	255	225	102.26	17	0
4	N5/1 TO N5/8	SIDE WALL RISER STUB	STEAM DRUM SHELL	8	114.3	SCH.XXS	17.12	SCH.40	225	-	-	255	225	102.26	17	0
5	N6/1 TO N6/6	FRONT & REAR WALL RISER STUB	STEAM DRUM SHELL	6	114.3	SCH.XXS	17.12	SCH.40	225	-	-	255	225	102.26	17	0
6	N9	FEED INLET (STUB + THERMAL SLEEVE)	STEAM DRUM SHELL	1	88.9	--	38.4	SCH.40	REF. DRG. -009-0001							125
7	N14/1 & N14/2	COMMON FEEDER NOZZLE	STEAM DRUM SHELL	2	273.1	SCH.160	28.58	SCH.120	250	-	-	382	352	230.22	28	70
8	N15	CBD NOZZLE	STEAM DRUM DISHED END	1	33.4	SCH.XXS	9.09	SCH.40	700	250	400	475	458	26.64	9	150
9	N16	IBD NOZZLE	STEAM DRUM SHELL	1	48.3	SCH.XXS	10.15	SCH.40	225	-	-	255	225	40.94	10	0
10	N18	AIR VENT	STEAM DRUM SHELL	1	48.3	SCH.XXS	10.15	SCH.40	225	-	-	255	225	40.94	10	0
11	N20	INSTRUMENT TEST/NITROGEN BLANKETING GAUGE NOZZLE	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	225	-	-	255	225	26.64	9	0
12	N22	LEVEL GAUGE NOZZLE (STEAM)	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	900	-	460	461	431	26.64	9	40
13	N23	LEVEL GAUGE NOZZLE (WATER)	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	900	-	460	461	431	26.64	9	40
14	N24	LEVEL GAUGE NOZZLE (STEAM)	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	900	-	460	461	431	26.64	9	40
15	N25	LEVEL GAUGE NOZZLE (WATER)	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	900	-	460	461	431	26.64	9	40
16	N26	LEVEL TRANSMITTER (STEAM)	STEAM DRUM DISHED END	1	33.4	SCH.XXS	9.09	SCH.40	700	350	305	459	429	26.64	9	125
17	N27	LEVEL TRANSMITTER (WATER)	STEAM DRUM DISHED END	1	33.4	SCH.XXS	9.09	SCH.40	700	350	305	459	429	26.64	9	125
18	N28	LEVEL TRANSMITTER (STEAM)	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	900	-	305	355	325	26.64	9	40
19	N29	LEVEL TRANSMITTER (WATER)	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	900	-	305	355	325	26.64	9	40
20	N30	LEVEL TRANSMITTER (STEAM)	STEAM DRUM DISHED END	1	33.4	SCH.XXS	9.09	SCH.40	700	350	305	459	429	26.64	9	125
21	N31	LEVEL TRANSMITTER (WATER)	STEAM DRUM DISHED END	1	33.4	SCH.XXS	9.09	SCH.40	700	350	305	459	429	26.64	9	125
22	N32	PRESSURE INDICATOR NOZZLE (LOCAL)	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	225	-	-	255	225	26.64	9	0
23	N33	PRESSURE TRANSMITTER NOZZLE	STEAM DRUM DISHED END	1	33.4	SCH.XXS	9.09	SCH.40	225	-	-	255	225	26.64	9	0
24	N34	REMOTE PRESSURE GAUGE NOZZLE (FLOOR)	STEAM DRUM DISHED END	1	33.4	SCH.XXS	9.09	SCH.40	225	-	-	255	225	26.64	9	0

CHAPTER : XII
IBR FIG : XII/85E B2

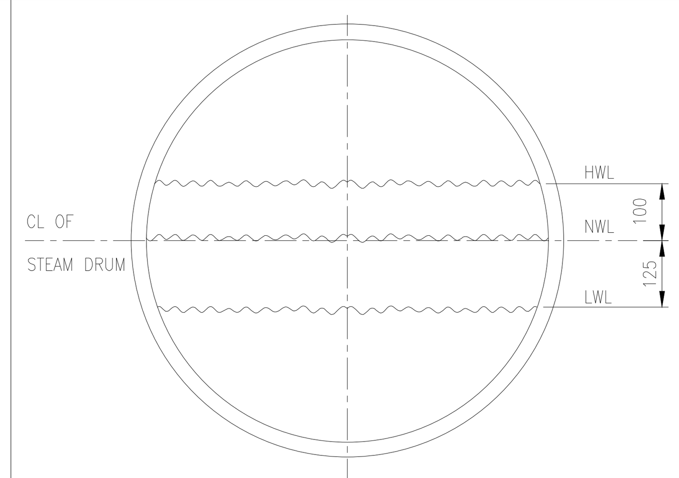


WELDING DETAIL OF NOZZLE TO DRUM
[FOR MARK Nos. N4/1,N4/2, N5/1 TO N5/8, N6/1 TO N6/6]

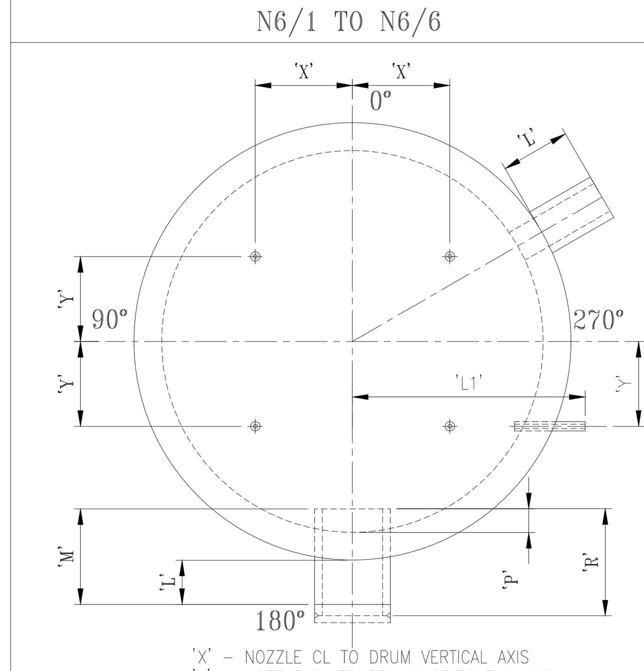
CHAPTER : XII
IBR FIG : XII/85E B2



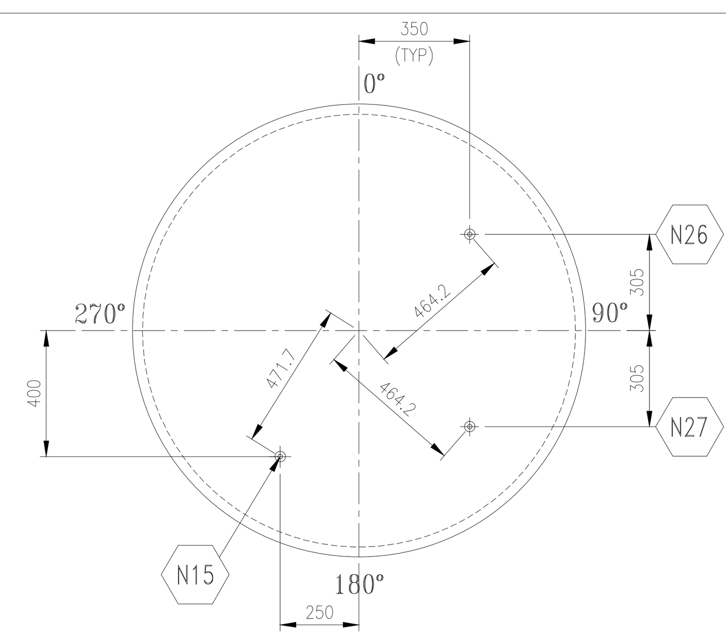
WELD DETAIL OF NOZZLE TO DRUM
MARK Nos. N1 & N2



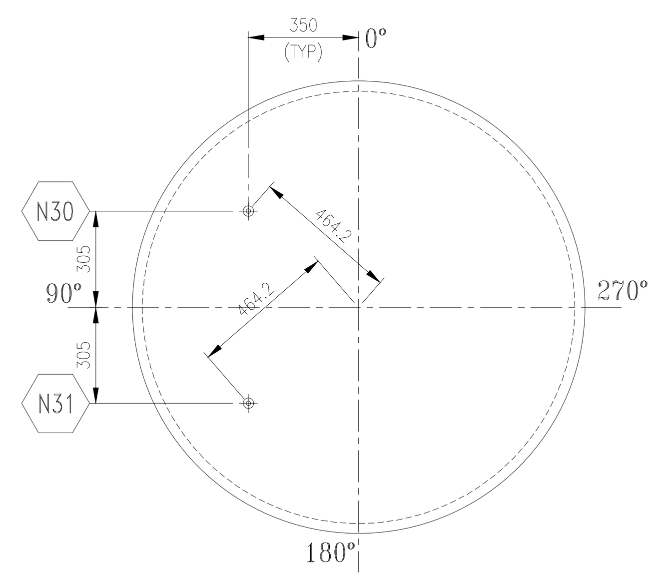
WATER LEVELS FOR STEAM DRUM



TYP DIM. IDENTIFICATION VIEW



DETAIL OF STEAM DRUM DISHED END PIPE NOZZLES
(TYP. FOR LHS)



DETAIL OF STEAM DRUM DISHED END PIPE NOZZLES
(TYP. FOR RHS)

REVN.	ZONE	DESCRIPTION	REVD. BY	CHKD. BY	APPD. BY	DATE

REVN.	ZONE	DESCRIPTION	REVD. BY	CHKD. BY	APPD. BY	DATE

CONSULTANT	--
CLIENT	M/s. JAMKHANDI SUGARS LTD (BAGALKOT, KARNATAKA)

DO NOT SCALE
IF IN DOUBT PLEASE ASK

FOR APPROVAL	NO	FOR INFORMATION	NO	FOR PRODUCTION	YES	FOR ERECTION	YES

RESPONSIBLE DEPT.	DRAWN BY	CHECKED BY	APPROVED BY	DATE
MECHANICAL	VENKAT.P	B.SARAVANAN/Y.SREEKANTH	R.ARUNKUMAR	25.11.2019

TOTAL WEIGHT IN kg.	PROJECT ID	ELEMENT ID	I.W.T. NO.
-	JB1160	50010100	6661
	YEAR : 2019	SHEET SIZE : A1	SCALE : 1:20

BOILERS
(formerly Isgec John Thompson)
Isgec Heavy Engineering Ltd.
Noida 201 301 (U.P.) India

SERVICE	SLOP FIRED TG BOILER GRITH SUPPORTED
TITLE	ARRANGEMENT AND DETAIL OF STEAM DRUM
DRAWING DOCUMENT ID	JB1160-50010100-007-0001
REVISION	00